

Take for an instance a 3D game that you play on your PC (the likes of Crysis, Need for Speed etc.) You could call some of them to have an immersive gameplay. Bring in a large display (say a 42-inch display) and you can get really lost into the world. In no time you start feeling that every single move in the game is actually happening with you in the real world. Every bullet you shoot and every turn you take feels like happening in real world though none are real. You start living in a reality that does not exist. That is the effect which scientists all around the world have been trying to enhance and as scientific research goes forward, a few things start coming to everyday technology.

A very popular game, Prototype's storyline was set in New York city. Locations, buildings, roads - all were designed into the game. During the gameplay, you run through the streets of the city, climb buildings. Let us assume you are not doing the gameplay on a single screen but a room with all walls as displays which make you believe that you are actually in the gameplay. Wouldn't that be really a great gameplay? You would be totally immersed.

Imagine an extension of this scenario. What if all those people in the game were real? By real we mean graphically generated models of real people right down to the clothes they were wearing. What if each one of those people were similarly plugged into this virtual universe, through virtual reality rooms of their own? The traffic could be created in a way to mimic actual traffic conditions on that day. So it the weather. Such a game wouldn't be so immersive that you would actually forget who you are and where you are?

Would you call that game a game? Or would you call it an application built around the concepts of 'augmented reality'? It's both. In a way Virtual Reality is a higher level of augmentation of reality - it is the peak. In the realms of Virtual Reality, it's not just the information that is added to reality but also control of that virtualised environment.

One of the main uses of virtual reality in the times we live is simulation programs that help train people. For instance, training Pilots. Allowing a newbie to take control of a fighter jet is extremely risky to both life and property and thus, flight simulation is one of the major uses of VR. For a VR flight simulator what could be the factors involved? Well, we'll need a 3D map of the reality and load it on a computer. We'll need to simulate the Aircraft's cockpit right down to the last knob and dial. The sounds that the plane produces right down to the beep, squeak and grid. We would need a pilot's seat and so on. This virtual environment will even need to react